

EnergyLens — Final Rapor Build Spec (tam görsel envanteri)

Her sayfada olması gereken HER öge: konum · tip · başlık (EN, sektör) · bağlı ölçü · durum. Bu PDF'i uygulayınca rapor final olur. ✓ =mevcut/çalışıyor · BAĞLA=yeniden bağla · YENİ=ekle · DÜZELT/BAŞLIK/DOĞRULA. Ölçü adları canlı modele göre.

0 · Önce script (final_master_install.cs)

TE2 → C# Script → **Open ikonu** ile aç (kopyala-yapıştırma) → **F5** → **Ctrl+S** → Refresh. Tüm ölçüleri + düzeltmeleri kurar (P3 status, IoT REMOVEFILTERS, öksüz V6/V2/IoT_CO2 sil+canlıya kur, specific-yield dedup, RTE ~90, CRREM bina-tipi, forecast band, climate EU1, feed-in tariff, realtime power, Date[Month Year] en-US, Selected Grain Label). Sonra aşağıdaki görselleri bağla/oluştur.

Çapraz: her zaman-serisi görselde **Sort axis → tarih/saat → Ascending**; metin alanlara Sort by column (hour_bucket→hour_of_day, Month Year→YearMonth). Aylık eksen = **Date[YearMonth]** (Türkçe ay adı çıkmaz). Time Grain etiketleri: Hourly/Daily/Monthly.

1 · Portfolio Overview

Portföy özeti — aylık + toggle. KPI'lar çalışıyor; tablo başlığı + EUI düzeltilecek.

Konum	Tip	Başlık (önerilen · EN)	Alan / Ölçü (tam ad) + durum
Üst slicer	Slicer	Date range + Building / City / Building Type / Country	✓ mevcut · Date dilimleyici + 4 alan slicer
KPI 1	Kart	Total Energy Consumption	✓ [Total Consumption kWh] · #,##0 kWh
KPI 2	Kart	Avg Energy Use Intensity (EUI)	DOĞRULA [Avg EUI kWh m2] — günlük 1,12 vs hedef 0,36 (bkz Veri böl.)
KPI 3	Kart	Active Anomalies	✓ [P3 Unresolved Count] / portföy anomali sayısı (mevcut)
KPI 4	Kart	Total Energy Cost (€)	✓ [Total Energy Cost EUR] · #,##0 €
KPI 5	Kart	Net Carbon Footprint (tCO ₂)	✓ [Net Carbon Footprint tCO2]
Kart R2 ×6	Kart	Last Week / WoW% / Last Month / MoM% / Last 7d Cost / Last 30d CO ₂	✓ 'Date' ölçüleri: [Card: Last Week ...], [Week over Week Consumption Pct], [Last Month kWh v14], [Month over Month Consumption Pct], [Last 7 Days Cost EUR], [Last 30 Days CO2 tCO2]
Orta sol	Tablo	Portfolio Building Summary	BAŞLIK düzelt (şu an yanlış: 'Monthly Energy Trend'). Sütun: building_name, country, building_type, energy_certificate, year_built, [Avg EUI kWh m2], [Total Energy Cost EUR]
Orta sağ	Bar	Consumption by Building — Current vs PY	✓ Eksen: silver_building_master[building_name] · [Total Consumption kWh] + [Total Energy PY kWh]
Alt sol	Çizgi	Portfolio Energy Consumption Trend	✓ toggle var · X: 'Time Grain' (default Monthly) · [TS Consumption kWh], [Total Energy PY kWh], [Energy Reduction Target kWh] · Sort axis ascending
Alt orta	Harita	Portfolio Map	✓ silver_building_master[city] (lat/long ya da şehir) · değer [Total Consumption kWh]
Alt sağ	Bar	Energy Use Intensity (EUI) Benchmark by Building	✓ Eksen building_name · [Avg EUI kWh m2] · sabit çizgi [EUI Target Numeric] / portföy medyan

2 · Building Deep-Dive ★

Operasyonel — toggle + 24h profil eklendi ✓. 2 kırık görsel kaldı.

Konum	Tip	Başlık (önerilen · EN)	Alan / Ölçü (tam ad) + durum
Üst slicer	Slicer	Date range + Building Name + Year Month	✓ mevcut
KPI ×6	Kart	Daily Consumption / Peak Demand / EUI / Solar Generation / Battery SoC / Carbon Intensity	✓ [Avg Daily Consumption kWh], [Peak Demand kW], [Avg EUI kWh m2], [Total Solar Generated kWh], [Avg Battery SoC Pct], [Avg Carbon Intensity kg m2]
R-A geniş	Çizgi	Consumption vs Outdoor Temperature (7-day rolling)	✓ toggle · X: 'Time Grain' (default Daily) · [TS Consumption kWh] + [TS Avg Temperature C] · Sort ascending
R-A sağ	Gauge	Energy Use Intensity (EUI)	✓ [Avg EUI kWh m2] · 0–1 skala
R-B sol	Çizgi	Average Daily Load Profile (24h)	✓ YENİ eklendi · X: gold_kpi_hourly[hour] (0–23) · [Profile Avg Demand by Hour kW]
R-B yanı	Kart	Base Load Ratio	✓ [Base Load Ratio Pct] · 0.0%
R-orta sol	Çizgi/Kolon	Solar Self-Consumption, Export & Grid Import	✓ toggle · X: 'Time Grain' · [TS Solar Self-Consumed kWh], [TS Net Grid kWh], [TS Solar Generated kWh]
R-orta sağ	Alan	Battery State of Charge (SoC) Trend	✓ toggle · X: 'Time Grain' · [TS Battery SoC Pct]
R-alt sol	Scatter	Consumption vs Heating Degree Days (HDD)	✓ X: [Total HDD] · Y: [Total Consumption kWh] · ay legend
R-alt sağ	Scatter	Consumption vs Cooling Degree Days (CDD)	✓ X: [Total CDD] · Y: [Total Consumption kWh]
(sol kırık ×2)	—	(eski 'Live Building Power' / 'Live HVAC Load')	DÜZELT KALDIR ya da gold_iot_realtime'a bağla ([Realtime Building Power kW], sensor_type filtreli)

3 · Anomaly Detection

Durum ölçüsü ✓. Başlık + Resolution Rate + sayım tutarlılığı.

Konum	Tip	Başlık (önerilen · EN)	Alan / ölçü (tam ad) + durum
Üst slicer	Slicer	Date range + Building / Severity / Anomaly Type	✓ mevcut
KPI ×5	Kart	Total / Critical Open / High / Unresolved / Active Anomaly Rate	✓ [P3 Total Anomaly Count], [P3 Critical Open Count], [P3 High Anomaly Count], [P3 Unresolved Count], [P3 Active Anomaly Rate Pct] · DOĞRULA: Unresolved>Total (bkz Veri)
Orta sol	Tablo	Operational Fault & Anomaly Register	✓ durum çalışıyor · building_name, severity, anomaly_type, detected_at, [P3 Anomaly Status], metric_value · BAŞLIK (opsiyonel)
Orta sağ	Yığılı kolon	Anomaly Trend by Severity	✓ X: Date[YearMonth] · count, legend severity
Alt sol	Bar	Anomalies by Type	✓ Eksen anomaly_type · count (SOLAR_UNDERPERFORM, AFTER_HOURS_WASTE, ...)
Alt sağ	Bar	Anomaly Share by Building (%)	✓ [P3 Building Anomaly Share Pct]
Alt kart ×4	Kart	Resolution Rate / Avg Days Open / High Priority Open / Top Risk Building	DÜZELT Resolution Rate '--'→0% format · ✓ [P3 Avg Days Open], [P3 High Priority Open Count], [P3 Top Risk Building Name]+[Score]

4 · Forecasting

Band + toggle ✓ (değiştirdiğin görsel buydu — doğru). Goal işareti.

Konum	Tip	Başlık (önerilen · EN)	Alan / Ölçü (tam ad) + durum
Üst slicer	Slicer	Date range + Building / Severity / Action Type	✓ mevcut
KPI ×5	Kart	7-Day / 30-Day Forecast / Savings Potential / CRITICAL Actions / Avg MAPE	✓ [Monthly Forecast kWh] türevleri + [Critical Recommendations]+[Avg Payback Years]; DOĞRULA 7-Day goal işareti
R-A geniş	Çizgi+alan	Consumption Forecast vs Actual (with confidence band)	✓ band+toggle · X: Date[YearMonth]/Time Grain (actuals) · [Monthly Actual kWh], [Monthly Forecast kWh], [Monthly PY kWh] + bant [Forecast Lower Bound kWh]...[Forecast Upper Bound kWh]
Sol kart ×2	Kart	Forecast Cost (30d) / Top Critical Saving	✓ mevcut
Orta alt	Scatter	Payback vs Savings	✓ X: payback_years · Y: annual_saving_eur · renk priority_label
Orta alt 2	Kolon+çizgi	Actions by Priority	✓ [Total Recommendations] / [Total Annual Saving EUR] by priority_label
Sağ alt	Bar	Forecast Accuracy by Building (MAPE)	✓ [model_mape_pct] (Excellent↔Poor); RELABEL 'in-sample fit'
Alt geniş	Tablo	Recommended Actions & Savings	✓ building, title_en, priority_label, [Total Annual Saving EUR], payback_years

5 · Occupancy Analysis

Zaten saatlik ✓. Occupied Hrs/Day imkansız değer.

Konum	Tip	Başlık (önerilen · EN)	Alan / ölçü (tam ad) + durum
Üst slicer	Slicer	Date range + Building / Profile Source / Day Name	✓ mevcut
KPI x4	Kart	Avg Occupancy / Peak Hour / After-Hours Share / Energy per Occupant	✓ [Avg Business Hours Occupancy Pct], peak hour, [Avg After Hours Occupancy Pct], [Energy per Occupant Goal kWh]
Orta geniş	Matrix	Weekly Occupancy Heatmap (7x24)	✓ satır day_of_week · sütun hour_of_day · [occupancy_probability] renk skalası
Alt sol	Çizgi	Hourly Occupancy — Weekday vs Weekend	✓ X: hour_of_day · [Avg Weekday Occupancy Pct], [Avg Weekend Occupancy Pct]
Alt sağ	Bar	Average Headcount by Building	✓ [Building Avg Headcount Business Hours] / peak
Sol kartlar	Kart	Ghost Load Risk / Est. Annual € Wasted / Occupied Hrs Day / Decision Insight	DÜZELT Occupied Hrs/Day 69,1 (>24, notebook 08) · ✓ [Ghost Load Risk Pct], [Ghost Load Annual Cost EUR], [Occupancy Top Insight]

6 · Sustainability / ESG ★

Renames + Scope2 loc/market ✓. CRREM script-fix. Veri büyüklüğü doğrulanmalı.

Konum	Tip	Başlık (önerilen · EN)	Alan / Ölçü (tam ad) + durum
Üst slicer	Slicer	Date + Country / Building Type / Certification / Building	✓ mevcut
KPI ×6	Kart	GHG Intensity / EPC Grade / CSRD Readiness / Solar Share / CO₂ Levy / CRREM Stranding Yr	DOĞRULA GHG 482 yüksek (Veri) · ✓ [EPC Class Area-Weighted], [CSRD Disclosure Readiness Pct], [CRREM Stranding Status]
Sol kart	Kart	EU Taxonomy Carbon Score	✓ [EU Taxonomy Carbon Score]
Orta sol	Çizgi	Carbon Intensity vs CRREM Pathway	✓ X: Date[Year] · [Carbon Intensity S1S2 kgCO ₂ m ² yr] + [CRREM Pathway 2C kgCO ₂ m ² yr] (script bina-tipi düzeltti)
Orta sağ	Tablo	Stranding Risk by Asset (CRREM)	✓ building, [Carbon Intensity S1S2...], [CRREM Pathway 2C...], [CRREM Stranding Status]
Alt sol	Yığılı kolon	GHG Emissions by Scope (1/2/3) per Building	✓ [Scope 1 tCO ₂], [Scope 2 Location tCO ₂], [Scope 3 Est tCO ₂]
Alt orta	Kolon	Scope 2 — Location vs Market (ESRS E1-6)	✓ scope2_location_tco2 / scope2_market_tco2
Alt sağ	Çizgi	Annual GHG Emissions Trend	✓ X: Date[Year] · [Total CO ₂ Emissions tCO ₂], [Net Carbon Footprint tCO ₂]
Sol alt	Harita	Portfolio Map (EU)	✓ şehir konumları
En alt	Kart/metin	Priority Actions & Incentives	✓ seçili bina önerisi (gold_recommendations)

7 · HVAC & Envelope

Renamed ✓. Kırık uptime görseli + DC CO₂.

Konum	Tip	Başlık (önerilen · EN)	Alan / Ölçü (tam ad) + durum
Üst slicer	Slicer	Date + Reno Priority / Year Month / Building / System Type	✓ mevcut
KPI x6	Kart	HVAC Share / HVAC Efficiency / Insulation / Heat Loss / HVAC CO ₂ / Buildings For Retrofit	✓ hvac_share_pct, hvac_efficiency_score, insulation_score, heat_loss_kwh_m2 (kolonlar); DOĞRULA HVAC CO ₂ (Veri)
Orta sol	Yığılı kolon	HVAC End-Use Breakdown — Heating/Cooling/Ventilation (modeled)	✓ X: Date[YearMonth] · heating/cooling/ventilation_kwh
Orta sağ	Tablo	HVAC System & Efficiency (COP/SCOP) by Building	✓ system_label, renovation_priority, cop_actual_avg/scop_rolling
Alt sol	Bar	HVAC CO ₂ — Scope 1 (Fossil) vs Scope 2 (Electric)	DOĞRULA Frankfurt DC 10.164.510 absürt (Veri, notebook 11)
Alt orta	Bar	Envelope Heat Loss Intensity (kWh/m ² ·yr)	✓ heat_loss_kwh_m2 + insulation_score
Alt sağ	Scatter	Retrofit Priority Matrix	✓ insulation_score × heat_loss · renk renovation_priority
Sol kart	Kart	EPC Compliance (x/10 buildings)	✓ mevcut
(sol kırık x2)	Matrix/Kart	Sensor Uptime by Zone / IoT CO ₂	BAĞLA script V2'yi gold_iot_daily_summary'e, IoT CO ₂ 'yi gold_iot_realtime'a kurdu → [V2 Sensor Uptime Pct]+[V2 Uptime Color Index] / [IoT CO ₂ Live]

8 · Real-Time IoT ★ (tüm görseller kırıldı — yeniden kur)

Öksüzler script ile temizlendi; ölçüler hazır. Görselleri bağla.

Konum	Tip	Başlık (önerilen · EN)	Alan / ölçü (tam ad) + durum
Üst slicer	Slicer	Date + Building Name + Building Type	✓ mevcut
KPI 1	Kart	Real-Time Building Power (kW)	BAĞLA [Realtime Building Power kW] (sensor_type='building_kwh' DOĞRULA)
KPI 2	Kart	Zone Comfort Compliance (EN 16798)	BAĞLA [IoT Zone Compliance Display] · renk [IoT Zone Compliance Color]
KPI 3	Kart	Indoor Air Quality — CO ₂ Level	BAĞLA [IoT CO2 Live]
KPI 4	Kart	Active Alerts & Estimated Cost	BAĞLA [IoT C4 Label v59] · renk [IoT C4 Color v59]
Orta sol	Çizgi	Building & HVAC Power — Last 24h	BAĞLA X: gold_iot_realtime[hour_of_day] · AVERAGE(reading_value_avg) · sensor_type building_kwh/hvac_kwh
Orta sağ	Matrix	Sensor Uptime by Zone	BAĞLA satır sensor_location/building · sütun sensor_type · [V2 Sensor Uptime Pct]
Alt sol	Bar	Zone Anomaly Count by Severity	BAĞLA gold_iot_realtime[anomaly_severity_window] count / [high/medium severity]
Alt sağ	Tablo	Fault Detection & Diagnostics (FDD) — Prioritised Findings	YENİ gold_iot_fdd · equipment, fault_code, severity, [IoT FDD Priority Band], [IoT FDD Row Confidence], [IoT FDD Row Cost Label], probable_cause · sort priority_score DESC
Kartlar	Kart	FDD Diagnoses / Max Priority / Avg Confidence / Energy Impact	YENİ [IoT FDD Diagnoses Today], [IoT FDD Max Priority], [IoT FDD Avg Confidence Pct], [IoT FDD Energy Impact kWh Today]

9 · Battery Dispatch & ROI ★

RTE + öksüz V6 script-fix. 2 görsel rebind + ROI.

Konum	Tip	Başlık (önerilen · EN)	Alan / Ölçü (tam ad) + durum
Üst slicer	Slicer	Building / Strategy / Battery Type + Date	✓ mevcut
KPI ×4	Kart	Annual Savings / Payback / CO₂ Avoided / Round-Trip Efficiency	✓ [C1 Annual Savings EUR], [C2 Payback Years], [C3 CO2 Avoided Tonnes Annual]; ✓ RTE [C4 Round Trip Efficiency Pct] script ~90'a düzeltti
Orta sol	Tablo	Battery Strategy Scenario Comparison (CAPEX/Payback/NPV/IRR)	DOĞRULA payback 230/IRR 450 (Veri, notebook 12) · building, strategy, capacity_kwh, total_capex_eur, [C1...], payback_years, npv, irr
Orta sağ	Çizgi (dual)	Monthly Savings & CO₂ Avoided	✓ X: Date[YearMonth] · [V4 Monthly Net Savings EUR], [V4 Monthly CO2 Avoided Kg]
R-B sol	Çizgi	24-Hour Battery Dispatch by Strategy	BAĞLA (V6 öksüzdü → script gold_battery_hourly_dispatch'e kurdu) · X: hour_label (Sort by hour_of_day) · [V6 Charge Rate], [V6 Discharge Rate], [V6 SoC Pct], [V6 Price Index]
R-B sağ	Matrix	Strategy Suitability by Building Type	✓ gold_strategy_fitness · [V1 Best Strategy For Building Type] + fitness_score renk
Kart	Kart	EU Battery Regulation (2023/1670) Compliance	BAĞLA [C5 EU Compliance Status Text]
Sağ panel	Kart	Battery Health — SoH & Cycles	✓ [V7 Current SoH Pct], [V7 Cycles Used Pct], [V7 Years Until Replacement]
Insight ×4	Kart	Strategy / Chemistry / Compliance / Next Action	✓ [I1 Strategy Recommendation Text], [I2 Chemistry Recommendation Text], [I3 Compliance Flag Text], [I4 Next Action Text]

10 · Solar (Full Showcase)

Generation aylık ✓ + feed-in ölçüsü ✓. Başlık + isim çıkışması (script) + 1 kırık.

Konum	Tip	Başlık (önerilen · EN)	Alan / Ölçü (tam ad) + durum
Üst slicer	Slicer	Building / Strategy / Battery Type + Date	✓ mevcut
KPI x6	Kart	Total Generation / Self-Consumption % / Avg PR / CO₂ Avoided / Renewable Rate / Installed Capacity	✓ [Total Solar Generated kWh], [Avg Self Consumption Rate Pct], [Avg Solar PR Pct], [Total CO2 Savings from Solar tCO2], [Renewable Energy Rate Pct], [Solar Capacity kWp]
R-A sol	Çizgi/alan	Solar Generation — Self-Consumed vs Exported	✓ aylık · X: Date[YearMonth] · [Solar Self Consumed kWh], [Total Solar Exported kWh]
R-A orta	Donut	Self-Consumption vs Export Split	✓ [Solar Self Consumed kWh] vs [Total Solar Exported kWh]
R-A sağ	Bar	PV Performance Ratio (PR) by Building (fault < 75)	✓ Eksen building_id · [Avg Solar PR Pct] · sabit çizgi 75
R-B sağ	Bar	Specific Yield by Building (kWh/kWp)	✓ script tekille düzeltir · [Avg Solar Specific Yield kWh kWp]
R-B sol	Çizgi	Live PV Power by Building (kW)	DÜZELT X: gold_iot_realtime[hour_bucket] Sort by hour_of_day · sensor_type=pv_ac_power · AVERAGE(reading_value_avg)
R-C sol	Tablo	PV Underperformance — Est. € Loss	✓ gold_iot_fdd fault=PV_UNDERPERFORMANCE · building_id, severity, occurrence_count, cost_eur_estimate
R-C alt	Tablo	Solar Investment Opportunities (INSTALL_SOLAR)	BAŞLIK düzelt (şu an 'PV Faults') · gold_recommendations action_type=INSTALL_SOLAR · country_code, priority_label, capex_eur, payback_years, npv_eur, co2_saving_kg, [Feed-in Tariff EUR kWh]

Veri / Notebook partisi (DAX değil — Fabric re-run; onay verince kod yazarım)

- GHG Intensity 482 / Frankfurt DC 2224 / Klinikum 751 kgCO2/m2/yr (Page 6) -> 09_ghg_scope: karbon yoğunluğu + alan bazı (conditioned vs gross).
- HVAC CO2 Frankfurt DC 10.164.510 (Page 7) -> 11_hvac_analytics: birim (kg/t) / DC outlier.
- Occupied Hrs/Day 69,1 >24 (Page 5) -> 08_occupancy_prediction: agregasyon.
- Battery payback 230 yıl / IRR %450 (Page 9) -> 12 (saat-bazlı fiyatlama, audit I3).
- Avg EUI 1,12/gün vs 0,36 hedef (Page 1/2) -> ölçü/03: yıllıklandırma ya da hedef günlüğe.
- Anomaly Unresolved 3K > Total 2,96K (Page 3) -> anomaly_detection / ölçüler: tutarlılık.
- Solar PV Share %1 (Page 6, hedef %40) -> gerçek mi / hesap mı.

Uygulama sırası

1. final_master_install.cs (Open+F5+Ctrl+S+Refresh)
2. Page 8 görsellerini sıfırdan bağla (KPI+24h+uptime+FDD)
3. Page 7/9 kırık görselleri rebind (V2 uptime / V6 dispatch / C5)
4. Page 2 kırık 2 görseli kaldır/bağla
5. Başlık: Page 1 tablo, Page 10 install tablosu
6. Sort axis ascending + YearMonth/en-US tüm trendlerde
7. Veri partisi (notebook fix + Fabric re-run)
8. Doğrulama turu